

AUTHBC — Running Results Summary

Co-optimizing encoding $\times$ authentication placement $\times$ signature scheme $\times$ batching
for blockchain-grade UAV telemetry over 802.11

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(living document — regenerated as each step completes)

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Abstract

This document tracks every measured result in the AUTHBC project and is updated at each step. All numbers trace to a frozen CSV in **results/** and are guarded by the reproduction gate (**make verify-frozen**). The headline is established: the co-optimized configuration cuts on-air authentication bytes by **75.0 %** versus the inline-CBOR baseline while holding verifiability $V \geq 0.95$ at loss $p = 0.05$ — criterion met. The hardware campaign has confirmed the one predicted anomaly (the signature-scheme ordering flips on ARM) and has shown the energy model’s assumed CPU power to be $4.1\times$ too high.

1 The problem

Telemetry records are tiny (45–190 B) but a signature is 64–96 B, so authentication overhead rivals the payload. Four knobs — record *encoding*, authentication *placement* (A inline, B self-batch, C relay-aggregate, D block), signature *scheme*, and *batch* size b — are coupled and must be co-optimized. The falsifiable criterion: the co-optimized configuration must cut on-air authentication bytes by $\geq 40\%$ versus inline-CBOR at $V \geq 0.95$, $p = 0.05$.

2 Software layer (frozen, reproducible)

2.1 E1 — overhead dominance (T1)

$\varphi = g/(s + g)$ is the fraction of on-air bytes that is pure signature ($g = 64$ B).

encoding	mean record s	φ
JSON	191.1 B	25.1 %
CBOR	66.25 B	49.1 %
MessagePack	65.16 B	49.6 %
delta	45.00 B	58.7 %

Reading. The better the compression, the *worse* the relative overhead: delta is the best encoder and leaves 59 % of the frame as signature. Compression alone cannot solve this — the motivation for the other three knobs.

2.2 E2 — batching cure (T2)

Folding b records under one signature gives $s + (g_a + H_f)/b$ per record; packing to the MTU bottoms out at $A = M/(M - H_f - g_a) = 1.0745$ at $M = 1500$. Measured: delta reaches $b_{\max} = 31$ and attains $A = 1.0745$ exactly; other encodings within 5 % (integer- b slack). Inline placement *never* amortizes its per-record signature — only the header shrinks.

2.3 E3 — loss frontier (T3)

Self-batch frames verify independently, so $V_B = 1 - p$, *flat in b* . A block signature spanning n frames needs all fragments: $V_D = (1 - p)^n$. Measured at $p = 0.10$, $b = 40$: $V_B = 0.90$ versus $V_D = 0.81 = 0.9^2$. This is the wall that stops over-aggregation.

2.4 E4 — scheme crossover (T4)

BLS aggregates but its pairing verify is $10.7\times$ costlier than Ed25519. Which wins depends on the powers only through $\kappa = P_r/P_c$, so the break-even κ^* is found without measuring watts. **Result: Ed25519 wins all 80 grid points**, $\min \kappa^* = 3.21$. At the accepted 96 B, BLS carries *more* bytes than Ed25519's 64 B on own traffic ($\kappa^* = \infty$); its aggregation pays off only for cross-signer relay traffic.

2.5 E5 — the headline (T5) **PASS**

config	enc	scheme	placement	auth B/rec	V
optimized	delta	Ed25519	B, $b=4$	26.00	0.95
A+CBOR (baseline)	cbor	Ed25519	A inline	104.0	0.95
A+JSON (naive)	json	Ed25519	A inline	104.0	0.95
D-over-agg	cbor	Ed25519	D, $b=40$	3.60	0.9025 FAIL

$$\text{cut} = \frac{104 - 26.0}{104} = \mathbf{75.0\%} \geq 40\% \Rightarrow \mathbf{PASS}$$

Hand-checked: $45.0 + 104/4 = 71.0$ B/record at $b=4$, freshness $4/20 = 200$ ms. **Freshness binds before the MTU does**: the byte-optimal batch is $b=31$ (96.8%) but sits at **1552 ms** of staleness, $6.2\times$ over the $D_{\max}=250$ ms bound docs/02 §7 requires the optimizer to enforce — which it computed and discarded until audit **F10**. Freshness is now a hard constraint and a fourth Pareto objective. The freshness-feasible batch obeys $b \lesssim \Delta D_{\max}$, independent of encoding and scheme. The resulting frontier ($b=1 \rightarrow 4$: 0/50/66.7/75% cut at 50/100/150/200 ms) is the co-design result; a single number is not. **The last row is the co-design argument in miniature**: over-aggregation is byte-competitive (3.60 B) yet *fails* $V \geq 0.95$ because $0.9025 = (1 - p)^2$. The byte-optimal answer is not the correct answer.

2.6 NS-3 channel validation

N	unicast vs ACK-Bianchi	broadcast vs naive reduction	broadcast vs Ma & Chen
5	+0.61%	−0.2%	−0.33%
20	+0.64%	+31.3%	−0.71%
50	− 2.85%	+ 1654.1%	+ 1.06%

Unicast validates the Bianchi DCF model to $+0.6--2.9\%$ (tightened from the previously reported $\pm 1.8-5.3\%$ by fixing F8, a $\sim 4.8\%$ goodput over-count in which the NS-3 sinks outlived the sources by 0.5 s on a 10 s window, and by using exact OFDM symbol timing). The middle column is our former in-house *reduction* of the unicast model to broadcast ($\tau = 2/(W+1)$); it fails not because of capture — measured at **0%** — but because it ignores the **backoff counter Consecutive Freeze Process**: with no ACK the contention window never doubles, so a station that has just transmitted may redraw backoff 0 and seize the medium one slot before any deferring station, whose counter is necessarily ≥ 1 . That is **Ma & Chen's published broadcast model** (IEEE Comm. Lett. 11(8):686–688, 2007; IEEE TVT 57(6):3757–3768, 2008), whose abstract warns that unicast models “cannot simply be reduced” to broadcast — exactly our error. Their closed form reproduces our measurement to $\leq 0.36\%$ (F9). **No novelty is claimed for the mechanism.**

3 Hardware layer (Raspberry Pi 4B, measured)

3.1 Cross-platform determinism

All **12/12** encoded-size metrics are byte-identical between x86 and ARM, as deterministic integer-only encoders require.

3.2 Timings and cycle normalisation

op	x86	RPi4	wall ratio	cycles ARM/x86
ed25519 sign	29.1 μ s	88.1	3.03 $\times$	1.16$\times$
ed25519 verify	95.0	259.5	2.73 $\times$	1.05$\times$
ecdsa sign	26.2	135.5	5.17 $\times$	1.98 $\times$
ecdsa verify	78.5	326.9	4.16 $\times$	1.60 $\times$
bls verify	1016.5	8611.8	8.47 $\times$	3.25 $\times$

Wall-clock ratio conflates clock speed with per-cycle efficiency. The clock floor is $4.7/1.8 = 2.61\times$. Normalising shows Ed25519 needs only 1.05 $\times$ the cycles on ARM (equally optimised on both ISAs) while BLS needs 3.25 $\times$ (heavily x86-tuned). **The docs/04 §1 “5–15 $\times$ ” wall-clock anchor was therefore withdrawn** as arithmetically unreachable and **restated per-cycle: expect 1–4 $\times$ the CYCLES on ARM** (clock- and platform-independent). All measured ops now sit inside that band.

3.3 F6 — the scheme ordering flips on ARM

platform	ECDSA verify	Ed25519 verify	winner
x86	78.5 μ s	95.0 μ s	ECDSA
RPi4	326.9	259.5	Ed25519

Three candidate causes were separated:

- **Library — ruled out.** Byte-identical versions on both (cryptography 45.0.7).
- **Our wrapper — real but not the cause.** Our ECDSA path does ASN.1 DER conversion in Python every op (Ed25519 does not): 1.05 μ s on x86, **12.58 μ s on ARM** ($\sim 12\times$ worse). Flipping the result would have required 64 $\times$. Core-only on ARM: ECDSA 312.4 versus Ed25519 263.3 — **Ed25519 still wins**.
- **Hardware — the operative cause**, but see §4: the *specific* microarchitectural mechanism is **not** established.

Consequence: on the real deployment platform the optimum selects **Ed25519** (the D2 architectural default). Both are 64B, so the auth-byte headline is **unchanged** by the scheme flip.

3.4 Measured per-op energy (RPi4, 60 s windows, 5 reps)

op	energy/op (μ J)	95 % CI	ΔP (W)
ed25519 sign	70.64	—	0.773
ed25519 verify	187.47	—	0.717
ecdsa sign	87.10	—	0.634
ecdsa verify	201.04	—	0.613
bls sign	1397.73	—	0.513
bls verify	4555.00	—	0.529
delta encode	36.92	—	0.726
cbor encode	40.43	—	0.753

Ed25519 is cheaper than ECDSA in energy as well as time on ARM (70.6 vs 87.3 μJ sign; 186.8 vs 203.5 verify), consistent with the timing flip. Independent cross-check against $\Delta P \times t_{op}$ (micro suite) agrees within 1–4.4 % for seven of eight ops; all eight ops now agree within **3.7 %** (BLS to 0.0 % and -0.1 %). The earlier 14.4 % `bls:verify` discrepancy was traced to warmup running *inside* the sync window and was fixed at source and **re-measured** (superseded data retained under `results/hw/energy/superseded/`). Measured p_{cpu} median **0.634 W** (range 0.513–0.773).

3.5 Reproducibility and energy

Two boards, independent runs: worst-case divergence **0.5 %** across all 16 timed ops. The energy chain validated end-to-end (12 windows $\rightarrow$ 12 segments) with two independent cross-checks (loop throughput versus micro suite within 5–6 %; single-core ΔP 0.728 W versus 4-core stress 0.690 W/core). **Measured $p_{cpu} \approx 0.73$ W against the nominal 3.0 W used in E5** — $4.1\times$ **too high**. The auth-byte headline is power-free and unaffected.

4 Open items and inconsistencies

item	status
F6 mechanism	Claim retracted. Masking ADX/BMI2 on x86 via <code>OPENSSL_ia32cap</code> did <i>not</i> slow ECDSA (61.1 $\rightarrow$ 56.8 μs ; ECDSA still won), so the ADX explanation is unsupported . What remains established: the ordering is ISA-dependent and Ed25519 is markedly more portable ($1.05\times$ versus $1.60\times$ cycles). The specific mechanism is an open question.
NS-3 broadcast	CLOSED (F9). PHY instrumentation shows capture contributes 0 % : all 1149 successful decodes at the monitored receiver came from single-transmitter busy periods, and an equal-power channel is byte-identical to the default. The cause is the Consecutive Freeze Process, and the corrected model is published (Ma & Chen 2007/2008): their closed form matches our NS-3 measurement to ≤ 0.36 % at every N . Our contribution here is validation at $W_0=16$ / 802.11a (they tested $W_0=32, 128$ at 1 Mb/s) and a direct PHY-trace measurement of the mechanism.
κ plausible band	E4 assumes $\kappa = P_r/P_c \lesssim 0.5$ from “radio below CPU power”. With p_{cpu} measured at 0.73 W (not 3.0 W), κ could approach or exceed 1. The E4 verdict is unchanged ($\min \kappa^* = 3.21$) but the stated band must be re-derived from measured powers.
p_{radio}	pending — needs the 2-node link run; both boards now healthy.
E4/E5 re-run	pending — deliberately deferred until both powers are measured, so the energy column stays internally consistent rather than half-measured.
5–15 $\times$ anchor	Needs restating per-cycle (decision required).

Provenance

Every number above traces to a frozen CSV under `results/`; derived artefacts are regenerated byte-identically by `make verify-frozen`, which fails loudly on any drift. Full audit: `docs/audits/p7.md`; decisions and their limitations: `docs/DECISIONS.md`.